Yidan Zhang 张一丹

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Seeking a research internship in multimodal LLMs and spatial intelligence.

Research Profile

Ph.D. student at CUHK-Shenzhen, advised by Prof. Xiaoguang Han. Researching multimodal models for dynamic physical worlds, focused on 4D spatial reasoning, long-video memory, and interactive agents. Experienced in large-scale multimodal data and video MLLM training.

Education

The Chinese University of Hong Kong, Shenzhen — Ph.D., Computer and Information Engineering Sep. 2024 – Present
School of Science and Engineering. Advisor: Prof. Xiaoguang Han.

The Chinese University of Hong Kong, Shenzhen — M.Phil., Computer and Information Engineering Sep. 2020 – Jul. 2023
Advisor: Prof. Xiaoguang Han. GPA 3.7/4.0.

Central China Normal University — B.Eng., Computer Science Sep. 2015 – Jun. 2019
Rank 4/196.

University of California, Berkeley — Exchange, Computer Science Aug. 2017 – Jan. 2018
GPA 3.7/4.0.

Experience

Huawei Noah's Ark Lab — Research Collaborator Sep. 2025 – Sep. 2026

- Collaborated with Dave Zhenyu Chen on MLLMs for 4D dynamic spatial reasoning and memory.
- Designed data/evaluation pipelines and static/dynamic spatial-memory modules for 4D scenes.

Selected Publications

T4D: Comprehending the Dynamic World with MLLMs Under review, ICLR 2027
Y. Zhang et al. **First author.**

- Builds an automated construction-and-verification pipeline for 4D spatial VQA, generating about 600K training examples and T4D-Bench (10 tasks, 3,250 expert-verified QA pairs across six dimensions).
- Introduces T4D, fusing explicit 3D instance trajectories from a frozen 4D geometry encoder with visual features via trajectory-aligned pre-training; reaches **61.0** accuracy on T4D-Bench (**+14.5** over same-data Qwen3-VL-4B) and comparable-scale SOTA.

VC-Agent: An Interactive Agent for Customized Video Dataset Collection ACM MM 2026
Y. Zhang et al. **First author.**

- Introduces VC-Agent, the first interactive MLLM agent for scalable customized video collection, using template-based acceptance and attribute-aware rejection to update its filtering policy from user feedback.
- Builds an end-to-end collection system with an easy-to-use Web UI, integrating video proposal, iterative criteria refinement, filtering, and automatic scale-up; user studies collect **335K** videos across eight datasets with only minutes of interaction per dataset and 84–92% judged quality.

MVImgNet: A Large-scale Dataset of Multi-view Images CVPR 2023
X. Yu*, M. Xu*, **Y. Zhang***, et al. **Co-first author.**

- Introduces MVImgNet, a large-scale real-world multi-view dataset with 6.5M frames from 219,188 videos across 238 classes, annotated with masks, cameras, and point clouds.
- Validates the data on radiance-field reconstruction, multi-view stereo, and view-consistent understanding, and derives MVP-Net with 87,200 point clouds for real-world 3D classification.

ReconViaGen: Towards Accurate Multi-view 3D Object Reconstruction via Generation ICLR 2026
J. Chang, C. Ye, et al., **Y. Zhang**, et al. Co-author.

- Identifies weak cross-view conditioning and unconstrained stochastic denoising as causes of inconsistency, and introduces the first 3D diffusion generator conditioned on reconstruction priors.
- Combines global and per-view reconstruction conditions with rendering-aware velocity compensation to align geometry and texture with input views; achieves SOTA on Dora-Bench and OmniObject3D.

Ongoing Research

Vision-TTT: Vision-Side Memory for Long-Video Spatial Intelligence In progress

- Introduces a vision-side fast-weight memory for long-video spatial intelligence, preserving 2D grids and VGGT geometry before visual tokens are flattened and compressing streaming spatial evidence into a fixed-size recurrent state.
- Trained on VSI-590K and 15K long-video counting examples; reaches **66.11** on VSI-Bench and **51.54** on ReVSI, and improves 10-minute spatial recall and counting by 10.0 points over Spatial-TTT-2B (**48.3**/**41.8** vs. 38.3/31.8).

Honors

2024 WAIC Youth Outstanding Paper Nomination Award
2018 National Scholarship
Academic Service: Reviewer for CVPR, ICCV, ACM MM, ICLR
CCF CAD&CG 2023 Open-source Dataset of the Year
Excellent Graduate, CCNU

Skills

Research. Spatial intelligence, 3D/4D vision, long-video understanding, multimodal LLMs. | **Technical.** Python, PyTorch, COLMAP. | **Languages.** Chinese (native); English (TOEFL 102).